

Hands-Free Object Selection via EOG Double-Blink Detection

A Live BCI Demonstration System

Summer Internship Report

Human-Computer Interface Lab

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1. Introduction

This report presents a working demo that allows a user to select an on-screen object without using their hands. The user first looks at a tile and performs a deliberate double blink to confirm the choice. The blink is detected in real time using electrooculography (EOG), where eye movement signals are captured through electrodes and streamed by an ESP32 microcontroller.

The system was developed as a proof-of-concept for a single subject (Aditya) rather than as a general-purpose product. The main goal was to build a complete end-to-end assistive interface, starting from signal acquisition and real-time detection to a responsive user interface. This report explains the overall design, implementation choices, challenges faced during live testing, and the final performance.

1.1 What the system does

- Acquires two-channel EOG at 256 Hz from an ESP32 over USB serial.
- Filters the signal to match the conditions the detector was tuned to.
- Detects a deliberate double-blink in real time and emits a single SELECT event.
- Highlights a grid tile (currently via trackpad/mouse; webcam gaze is a future upgrade) and selects the highlighted tile on SELECT.
- Shows a live signal monitor so contact quality and detected blinks are visible during operation.

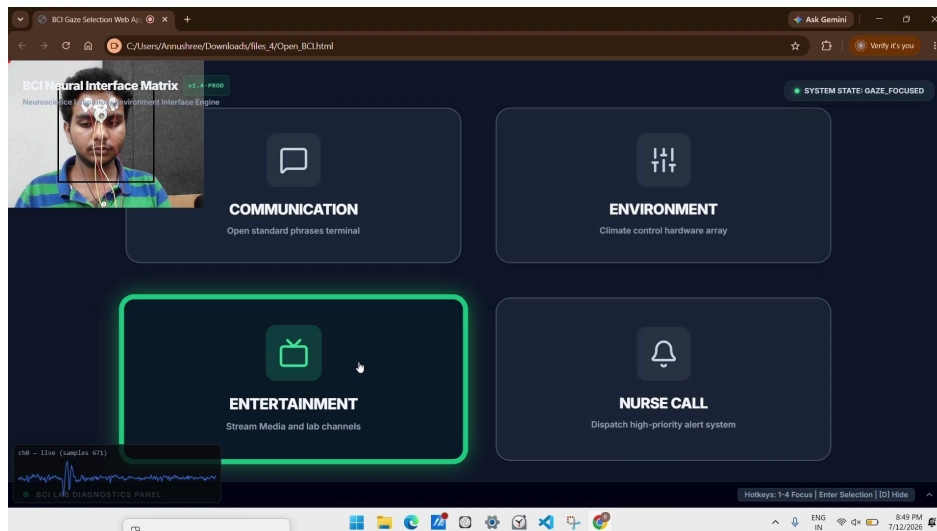


Figure 1. The live interface: a 2x2 selection grid with a highlighted tile (green), the subject's webcam preview (top-left), and the live *ch0* signal monitor (bottom-left).

2. Overall System Architecture

The system is divided into two main parts: a Python backend that handles hardware communication and blink detection, and a browser frontend that manages the user interface. Both parts communicate through a lightweight one-way event channel, keeping the signal-processing pipeline independent from UI rendering.

2.1 Data Pipeline

ESP32 (256 Hz, 2ch EOG)

- serial reader thread (bulk read, parse, queue)
- worker thread (filter → detect → optional log)
- SELECT event → SSE bridge → browser → grid selects tile

The serial reader thread is responsible only for reading incoming bytes, converting them into samples, and placing them into a bounded queue. The worker thread then filters the signal, runs the blink detector, and optionally stores logs. Separating these tasks prevents heavy processing from affecting data acquisition timing.

2.2 Components

Component	Role
ESP32 + 2-ch EOG electrodes	Signal acquisition at 256 Hz streams timestamped samples over USB serial.
Preprocessing (CollectorFilter)	Reproduces the exact filter the training & tuning data used (0.5 Hz high-pass → 50 Hz notch → 45 Hz low-pass), stateful and per-channel.
Rule Detector (RuleBasedBlinkDetector)	Real-time double-blink detection on ch0. No model or training data required.
Live Blink Detector(EOG_NET)	Optional ML alternative: a CNN + Transformer classifier. Selectable but not the default.
bci_bridge	Server-Sent-Events channel that pushes SELECT (and monitor) events to the browser.
Open_BCI.html + integration script	The React grid UI highlights tiles and selects the highlighted one on SELECT.
live_demo.py	The orchestrator: threads, filter, detector, bridge, and optional logging.

3. Signal Processing Pipeline

3.1 Electrode Placement and Effect

The electrode placement remained the same for every recording collected during this project. Positive electrodes were placed at Fp1 and Fp2, the reference electrode behind the ear (mastoid), and the ground electrode on the forehead. This forms a standard frontal EOG montage.

This placement is close to ideal *for blink detection*. Fp1 and Fp2 are exactly where blink potentials are largest in EEG, the eyelid sweeping the cornea, together with the eyeball rolling upward during a blink (Bell's phenomenon), drives a large deflection directly under these electrodes. This is why a well-contacted channel produces big, easily thresholded blink excursions. Two further consequences follow from the montage: the two channels are largely *redundant* (a blink is bilateral, so Fp1 and Fp2 see nearly the same thing, which is why the single-channel detector is sufficient), and the blink polarity is consistent across sessions (observed negative on ch0, which the detector is configured to match).

A key limitation of this montage is its sensitivity to vertical gaze, which accounts for the observed behaviour. Because the Fp1 and Fp2 electrodes sit within the corneo-retinal dipole's field, they capture vertical eye

movements as well as blinks. Specifically, looking upward shifts the dipole toward these electrodes, producing a signal deflection similar to a blink, while looking downward shifts it in the opposite direction.

3.2 Preprocessing

Since EOG blinks primarily contain low-frequency information, the incoming signal is filtered using the same pipeline used to prepare the detector. The signal passes through a 0.5 Hz high-pass filter and a 50 Hz notch filter to remove power and match the offline recordings.

3.3 Rule-Based Blink Detector

The main detector is intentionally lightweight. It does not require a trained model, stored normalisation values, or training data. It works only on channel ch0, making it reliable even if the second electrode has poor contact. The detection process consists of three stages:

1. **Adaptive noise floor.** A robust baseline (median) and noise scale (median absolute deviation) are computed over a rolling 10-second window. Because MAD ignores the brief, large blink samples, the noise estimate reflects genuine rest noise and adapts automatically to the session.
2. **Blink detection.** A blink is a contiguous excursion beyond 7 standard deviations of the baseline (negative-going, matching the subject's montage). Detecting the whole excursion as one event rather than each threshold crossing naturally avoids counting a single biphasic blink twice. Each blink's peak amplitude is recorded.
3. **Double-blink confirmation.** Two blinks whose separation falls in a permitted window and whose peak amplitudes are comparable are confirmed as a double-blink. A debounce period then prevents one gesture from firing twice. r-line noise, and finally a 45 Hz low-pass filter. The filters operate sample-by-sample so that the live signal closely

3.4 Importance of Amplitude Matching

Amplitude comparison turned out to be one of the most important additions to the detector. During testing, single blinks often produced small ringing artefacts after filtering, which looked similar to the second blink of a fast double blink. Looking only at timing could therefore lead to false detections.

The difference becomes clear when amplitude is also considered. A real double blink contains two peaks of similar size, while a ringing artefact has a much smaller second peak. Comparing the peak amplitudes allows the detector to reject ringing while still accepting genuine fast double blinks.

3.5 Deep Learning Detector

A machine-learning-based detector is also available and can be enabled through a configuration option. It uses a CNN + Transformer architecture (EOG_NET) on fixed 3-second windows. Although it performs well, it requires both channels, normalisation statistics, and a full analysis window, introducing additional latency(time delay). For the live demo, the simpler rule-based detector was more practical and reliable.

4. User Interface

The frontend is a browser-based React application that displays a simple 2×2 grid with large labelled tiles such as Communication, Environment, Entertainment, and Nurse Call. It listens for two events: highlighting a tile and confirming the highlighted selection.

- **Highlighting** is currently driven by the trackpad/mouse. Webcam gaze tracking was implemented but is coarse and calibration-sensitive, so it is treated as a deferred enhancement rather than a dependency (see Section 6).
- **Selection** is driven by the real double-blink detector: a confirmed double-blink is delivered to the browser and selects the highlighted tile.

A small on-screen signal monitor renders the live ch0 trace and flashes on each detected blink and selection. It reads the same event channel as the grid and adds no measurable cost to the detection path, so contact quality and detector activity are visible at a glance during operation.

5. Development Challenges and Fixes

Most of the work during development went into making the complete system reliable on live hardware rather than improving the detector itself. Each issue was analysed using recorded logs together with the ESP32 timestamps, confirming that sampling remained stable at 256 Hz and that the problems originated on the host side.

Symptom	Root cause	Fix
Constant false firing at rest	Electrodes not connected → floating inputs saturate the ADC.	Connect and seat electrodes; verify on the monitor before use.
No response at all	Second channel (ch1) is dead or railing; signal far from training scale.	Reseat/repair ch1; use the ch0-only rule detector.
Fires only when looking up	EOG blink shape is entangled with gaze direction; the detector saw only one gaze condition.	Keep demo gaze matched to the tuning condition; gaze-varied retraining noted as future work.
5–7 second lag, growing	The serial reader read one line at a time and fell behind; a debug print made it worse.	Bulk-read the whole buffer and split into lines; remove per-iteration printing; bound the queue.
Fires on single blinks	Filter ring created false second blinks at the same timing as real fast doubles.	Amplitude-comparison guard (Section 3.3).
Zero fires after tightening	The timing window widened to reject rings had also rejected the subject's fast doubles.	Loosen timing; rely on amplitude to reject rings instead.

One clear observation throughout development was that electrode contact had the biggest impact on performance. The same detector could work very well in one recording and poorly in another simply because of contact quality. Checking the live signal before starting a session became the most important step.

6. Results and Limitations

6.1 Observed behaviour

With good electrode contact, the rule-based detector consistently identifies deliberate double blinks in real time and selects the highlighted tile. The overall delay is roughly equal to the time needed to perform the double-blink gesture itself. On the labelled reference recording, all intended double blinks were detected without false triggers during rest.

6.2 Honest limitations

- **Single subject:** All tuning is specific to one person and electrode placement. Generalisation to other users is a separate, harder problem and was not a goal.
- **Contact-sensitive:** Reliability depends heavily on stable ch0 contact. Poor or drifting contact degrades detection; this is a hardware property, not a software one.
- **Gaze-condition dependence:** Because EOG mixes blink and gaze, the detector is tuned to one gaze condition. A gaze-invariant detector requires training data collected across gaze directions.
- **Timing quantified by inference, not labels:** Live parameter tuning was guided by unlabelled session logs. A short-cued (labelled) recording would allow exact recall and false-positive figures to be reported rather than inferred.

6.3 Future work

- Record a short cued calibration session to fix detector parameters to measured values and report exact accuracy.
- Retrain across gaze directions for a gaze-invariant detector, enabling reliable webcam-gaze highlighting.
- Integrate webcam gaze tracking for the highlight once the detector is gaze-invariant.
- Improve ch1 electrode robustness to enable the two-channel ML detector as a cross-check.

7. Conclusion

This project demonstrates a complete hands-free selection system using two-channel EOG acquired through an ESP32, real-time double-blink detection, and a responsive browser interface. The final detector is a simple rule-based approach that uses blink amplitude to distinguish genuine double blinks from filter ringing artefacts. The biggest lesson from the project is that stable signal acquisition and good electrode contact are more important than using a more complex detection algorithm.